

Universal Project Deployment Guide

Standard Operating Procedure: Pushing Any Codebase to GitHub & Live Deployment on Vercel

UNIVERSAL REFERENCE MANUAL

1 Security & Cleanliness: Setup `.gitignore`

Before initializing or staging files, always prevent secret keys, credentials, and bulky dependency folders from leaking into public repositories.

Create a file named `.gitignore` in your project root directory:

```
# Dependencies
node_modules/

# Production build output
dist/
build/
.output/

# Environment secrets & API keys (CRITICAL)
.env
.env.*
!.env.example

# System & IDE artifacts
.DS_Store
Thumbs.db
.vscode/
.idea/
```

Important: Always commit an empty `.env.example` with placeholder variable names so collaborators know which environment keys are required without exposing real credentials.

2 Pushing Code to GitHub

Open your command line / terminal inside the project root folder and execute the standard sequence:

```
# 1. Initialize local repository and set default branch
git init
git branch -M main

# 2. Configure Git author identity (if not already set globally)
git config user.name "YourGitHubUsername"
git config user.email "your-email@users.noreply.github.com"

# 3. Stage all files and verify clean tree
git add .
git status

# 4. Commit files
git commit -m "Initial commit: Project source and architecture"

# 5. Link to your empty GitHub repository and push
git remote add origin https://github.com/<your-username>/<your-repo-name>.git
git push -u origin main
```

3 Configure SPA Routing for Vercel (`vercel.json`)

For modern Single Page Applications (Vite, React, Vue, Svelte), create a `vercel.json` in the project root to prevent **404 Not Found** errors when refreshing sub-routes:

```
{
  "framework": "vite",
  "buildCommand": "npm run build",
  "outputDirectory": "dist",
  "cleanUrls": true,
  "rewrites": [
    {
      "source": "/(.*)",
      "destination": "/index.html"
    }
  ]
}
```

Tip for Pure Static HTML: If your project uses raw HTML without a build tool like Vite, omit `buildCommand` and set `outputDirectory` to `"."`.

4 Connecting to Vercel

- 1. Log in to your account at vercel.com.
- 2. Click "Add New..." → "Project".
- 3. Under **Import Git Repository**, choose your repository and click **Import**.
- 4. Vercel automatically detects the framework (e.g. **Vite**) with build command `npm run build` and output directory `dist`.
- 5. If using environment variables (e.g., Database connection or API tokens), add them under **Environment Variables**.
- 6. Click **Deploy**.

5 Ongoing Development & Continuous Deployment (CI/CD)

Once connected, you never need to manually upload files again. Any push to the `main` branch triggers an automatic production build:

```
git add .
git commit -m "Add new feature or layout improvements"
git push origin main
```

Vercel detects the push, runs the build command, and publishes the new version live in seconds.

6 Quick Reference Cheat Sheet

Workflow Step	Command / Action	Notes
Local Preview	<code>npm run dev</code> or <code>npm run preview</code>	Test locally before pushing
Test Production Build	<code>npm run build</code>	Verifies no build errors in <code>dist/</code>
Stage Changes	<code>git add .</code>	Stages all modified files
Commit Changes	<code>git commit -m "Message"</code>	Creates version snapshot
Deploy Live	<code>git push origin main</code>	Vercel deploys immediately